

Ch 3. Quantum Mechanics.

- Nobel prize.

Max Planck (1918), Albert Einstein (1921), Niels Bohr (1922),
Werner Heisenberg (1932), Schrödinger, Dirac (1933), Pauli (1945),
Max Born (1954), Richard Feynman, Julian Schwinger, Sin-Itiro Tomonaga
(1965).

Alain Aspect, John Clauser, Anton Zeilinger ~~†~~ (2022).

: Experimentally verified "entanglement".

"Closed physical systems can exist in a superposition of many possible states."

qubit: points on unit sphere with radius 1 (3D) $\rightarrow$ Bloch sphere.
Describe quantum states using density operators.

§ 3.1 State spaces.

Postulate 3.1.1 (State space postulate)

A closed physical system is associated with a Hilbert space, called the "state space" of the system. The system at a particular time is completely described by a unit vector in its state space, called the "state vector" or "state" of the physical system.

Also called "wave function".

Possible to construct good approx. as being closed.

3.1.2 Quantum bits

Basic unit of information.

State space $\mathbb{H}_1$ (Def 2.1.2), 2^1 complex Hilbert space $\overbrace{(|0\rangle, |1\rangle)}^{\text{computational basis}}$.

$$|\varphi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle, \quad \alpha_0, \alpha_1 \in \mathbb{C}, \quad \|\varphi\|^2 = |\alpha_0|^2 + |\alpha_1|^2 = 1.$$

$\uparrow$
superposition of $|0\rangle$ & $|1\rangle$. α_0, α_1 : amplitude for $|0\rangle, |1\rangle$.

Def 3.1.2 Let $k \in \mathbb{N}$, $(|\varphi_0\rangle, \dots, |\varphi_{k-1}\rangle) \in H^k$ be linearly indep.

$|\varphi\rangle \in H$, $|\varphi\rangle = \alpha_0 |\varphi_0\rangle + \dots + \alpha_{k-1} |\varphi_{k-1}\rangle$, $\alpha_i \in \mathbb{C}$.

For each i , α_i = amplitude of $|\varphi\rangle$ for the state $|\varphi_i\rangle$.

physical meaning: "probability". Ex $|\varphi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$
 $\|\varphi\|^2 = 1 = \frac{1}{2} + \frac{1}{2}$
↑ ↑
prob prob

3.1.3 Spherical coordinates

"Bloch sphere"

$\vec{p} = (x, y, z) \in \mathbb{R}^3$. Another basis B : $\vec{p}_B = B^{-1} \vec{p}$.

Ex $\vec{p} = (1, 1, 1)$, $B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \Rightarrow B^{-1} \vec{p} = (1, -1, -1)$

Def 3.1.5 $\vec{p} = (p_x, p_y, p_z)$, $\vec{g} = (g_x, g_y, g_z) \in \mathbb{R}^3$

(1) $\langle \vec{p} | \vec{g} \rangle = \vec{p} \cdot \vec{g} = p_x g_x + p_y g_y + p_z g_z$

(2) ~~Euclidean~~ $\|\vec{p}\| = \sqrt{\langle \vec{p} | \vec{p} \rangle}$

(3) $\|\vec{p}\| = 1$: unit vector

(4) $\vec{p}, \vec{g}$: ortho $\Leftrightarrow \langle \vec{p} | \vec{g} \rangle = 0$

orthogonal basis, orthonormal basis...

Prop 3.1.6 Hermitian, Positive definite.

Prop 3.1.12 $\vec{p} \in \mathbb{R}^3$, $\vec{p} \neq 0$. $\vec{p} = \|\vec{p}\| (\cos\phi \sin\theta, \sin\phi \sin\theta, \cos\theta)$
 $0 \leq \theta \leq \pi$, $\begin{cases} \phi = 0 & \text{if } \theta \in \{0, \pi\} \\ 0 < \phi < 2\pi & \text{o.w.} \end{cases}$

3.1.4 The Bloch Sphere.

Def 3.1.14 Bloch sphere: $\{\vec{p} \in \mathbb{R}^3 : \|\vec{p}\| = 1\}$.

single qubit states can be represented by 3 real numbers.

Prop 3.1.18 $|\psi\rangle \in H$: single-qubit state. Then

$$|\psi\rangle = e^{i\gamma} \left(\cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle \right)$$

$$\begin{aligned} 0 &\leq \theta \leq \pi \\ 0 &\leq \gamma, \phi < 2\pi \\ \theta \in \{0, \pi\} &\Rightarrow \phi = 0 \end{aligned}$$

Bloch sphere
↓
 ψ

$$\forall |\psi\rangle \in H, \rightarrow \vec{p}(\psi) = (1, \theta(\psi), \phi(\psi))$$

$$((\sin \theta(\psi) \cos \phi(\psi), \sin \theta(\psi) \sin \phi(\psi), \cos \theta(\psi)))$$

$$\vec{p} = (1, \theta, \phi) \rightarrow |\psi(\vec{p})\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle$$

$r?$

Def 3.1.22 Let $|e\rangle, |\psi\rangle \in H$, $|\psi\rangle = e^{ir}|\psi\rangle$.

We say $|e\rangle$ and $|\psi\rangle$ are equal up to the global phase factor e^{ir} .

This gives an equivalence relation, we denote the ^{equivalence} class of $|\psi\rangle$ by $[\psi]$.

(Thm 3.4.4 $\Rightarrow$ δ has no impact on measurement outcome)

Thm 3.1.25 S_1 = quantum states in H , R_1 = equiv rel.

$$S_1/R_1 \longleftrightarrow \{\vec{p} \in \mathbb{R}^3 : \|\vec{p}\| = 1\}$$

$$[\psi] \longmapsto \vec{p}(\psi)$$

$$[\psi(\vec{p})] \longleftarrow \vec{p}$$

Ex (2.4.7) Pauli operators X, Y, Z

orthonormal eigenbases $(|x_+\rangle, |x_-\rangle) = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}, \frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$

$$(|y_+\rangle, |y_-\rangle) = \left(\frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \frac{|0\rangle - i|1\rangle}{\sqrt{2}}\right)$$

(3.1.19): type

$$(|z_+\rangle, |z_-\rangle) = (|0\rangle, |1\rangle)$$

$$|z_+\rangle = |0\rangle = e^{i \cdot 0} \left(\cos \frac{0}{2} |0\rangle + e^{i0} \sin \frac{0}{2} |1\rangle \right) \Rightarrow (1, 0, 0) \Rightarrow (0, 0, 1)$$

$$|z_-\rangle = |1\rangle = e^{i \cdot 0} \left(\cos \frac{\pi}{2} |0\rangle + e^{i0} \sin \frac{\pi}{2} |1\rangle \right) \Rightarrow (1, \pi, 0) \Rightarrow (0, 0, -1)$$

$$|y_-\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}} = e^{i \cdot 0} \left(\cos \frac{\pi}{4} |0\rangle + e^{i \cdot \frac{3\pi}{4}} \sin \frac{\pi}{4} |1\rangle \right) \Rightarrow (1, \frac{\pi}{2}, \frac{3\pi}{4}) \Rightarrow (0, -1, 0)$$

3.1.5 Quantum registers.

A quantum system comprising n qubits is called an "n-qubit quantum register".

$$|\psi\rangle = \sum_{\vec{b} \in \{0,1\}^n} \alpha_{\vec{b}} |\vec{b}\rangle, \quad \sum |\alpha_{\vec{b}}|^2 = 1$$

↑
n-qubit state

↑
superposition of basic elements

ex $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{i}{\sqrt{2}}|11\rangle = \frac{1}{\sqrt{2}}|0\rangle_2 - \frac{i}{\sqrt{2}}|3\rangle_2$

$|\alpha_{\vec{b}}|^2$: prob $\vec{b}$ observed. After measurement with outcome $\vec{b}$, the quantum register's state becomes $|\vec{b}\rangle$.

§3.2 State spaces of composite systems.

Postulate 3.2.1 (Composite Systems Postulate)

The state space of the composition of finitely many physical systems is the tensor product of the state spaces of the component systems. Moreover, if we have systems numbered 0 through $m-1$ and if system i is in the state $|\varphi_i\rangle$ for $0 \leq i < m$, then the composite system is in state $|\varphi_0\rangle \otimes \dots \otimes |\varphi_{m-1}\rangle$.

$$H = H_1 = H_{n_0} \otimes \dots \otimes H_{n_{m-1}}, \quad n = \sum_{i=0}^{m-1} n_i.$$

3.2.2 Entangled States. (Recall Def 2.5.20).

Ex 3.2.2 $H_2 = H_1 \otimes H_1$.

H_2 contains the composite state $|\varphi\rangle = |\varphi_0\rangle \otimes |\varphi_1\rangle$.

As we saw in Ex 2.5.22, the Bell state $|\varphi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ cannot be written in this form, It is therefore entangled.

Def 3.2.3 A state of the composition of two physical systems is called "entangled" if it cannot be written as the tensor product of states of the component systems. Otherwise, this state is called separable or nonentangled.

Thm 3.2.4 The state of the composition of two quantum systems is separable iff its Schmidt rank is 1, and it is entangled iff its Schmidt rank is greater than 1.

§3.3 Time evolution. (How does the state change over time?)

Postulate 3.3.1 (Evolution Postulate) The evolution of a closed quantum system is described by a unitary transformation. More precisely, if $t, t' \in \mathbb{R}$, $t < t'$, then the state $|\varphi'\rangle$ of the system at time t' is obtained from the state $|\varphi\rangle$ of the system at time t as $|\varphi'\rangle = U|\varphi\rangle$ where U is a unitary operator on the state space of the system that depends only on t & t' .

3.3.2

Quantum gates

input state $\xrightarrow{\text{unitary transformation}}$ output state

Hadamard gate (or Hadamard operator)

$$H: H_1 \rightarrow H_1, \quad |0\rangle \mapsto |x_+\rangle, \quad |1\rangle \mapsto |x_-\rangle, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

(Ex) H : Hermitian unitary involution, i.e. $H^* = H = H^{-1}$. H

(Ex) CNOT gate

$$\text{CNOT}: H_2 \rightarrow H_2, \quad |c\rangle|t\rangle \mapsto |c\rangle X^c|t\rangle$$

$\uparrow$ $\uparrow$
 control qubit target qubit

Applies the Pauli X operator to $|t\rangle$ if $|c\rangle = |1\rangle$

$$|00\rangle \mapsto |00\rangle, \quad |01\rangle \mapsto |01\rangle, \quad |10\rangle \mapsto |11\rangle, \quad |11\rangle \mapsto |10\rangle$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



$$\text{CNOT}^* = \text{CNOT} = \text{CNOT}^{-1}, \quad \text{CNOT}^2 = I$$

Does CNOT gate never changes the control qubit?

$|x_+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$
 $|x_-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$

(Ex) $\text{CNOT}|x_+\rangle|x_-\rangle = \frac{\text{CNOT}|0\rangle|0\rangle - \text{CNOT}|0\rangle|1\rangle + \text{CNOT}|1\rangle|0\rangle - \text{CNOT}|1\rangle|1\rangle}{2}$

$$= \frac{|0\rangle|0\rangle - |0\rangle|1\rangle + |1\rangle|1\rangle - |1\rangle|0\rangle}{2} = |x_-\rangle|x_-\rangle$$

: CNOT changes the control qubit but not the target qubit.

3.3.3

Composition of operators

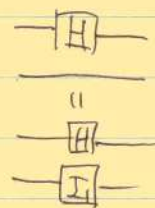
 $m \in \mathbb{N}$, m quantum systems $H(0), \dots, H(m-1)$

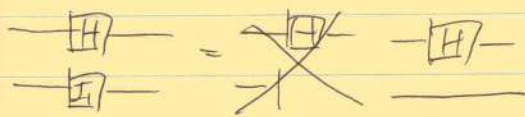
$$H = H(0) \otimes \dots \otimes H(m-1)$$

 $f_0, \dots, f_{m-1}$: linear operators on $H(0), \dots, H(m-1)$

$$f := f_0 \otimes \dots \otimes f_{m-1}$$

$$f_j \text{ on } H(j) = I_0 \otimes \dots \otimes I_{j-1} \otimes f_j \otimes I_{j+1} \otimes \dots \otimes I_{m-1}$$

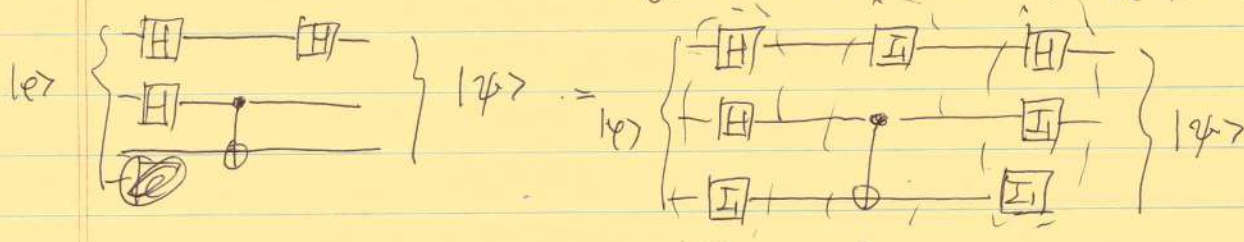




$$\begin{aligned} |00\rangle &\mapsto \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \\ |01\rangle &\mapsto \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) \\ |10\rangle &\mapsto \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) \\ |11\rangle &\mapsto \frac{1}{\sqrt{2}}(|01\rangle - |11\rangle) \end{aligned}$$

$$H \otimes I_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

§ 3.4 Quantum Circuits.



$$U = U_2 \circ U_1 \circ U_0$$

$$\begin{aligned} |000\rangle &\xrightarrow{U_0} \frac{1}{\sqrt{4}}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle)|0\rangle \\ &= \frac{1}{\sqrt{4}}(|000\rangle + |010\rangle + |100\rangle + |110\rangle) \end{aligned}$$

$$\xrightarrow{U_1} \frac{1}{\sqrt{4}}(|000\rangle + |011\rangle + |100\rangle + |111\rangle)$$

$$= \frac{1}{\sqrt{4}}(|0\rangle + |1\rangle)(|00\rangle + |11\rangle)$$

$$\xrightarrow{U_2} \frac{1}{\sqrt{2}}|0\rangle(|00\rangle + |11\rangle)$$

$$= \frac{1}{\sqrt{2}}(|000\rangle + |011\rangle)$$

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|1\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\therefore |000\rangle \xrightarrow{U} \frac{1}{\sqrt{2}}(|000\rangle + |011\rangle)$$

§ 3.4 Measurements

When a device makes a measurement, the state of the quantum system "collapses" irreversibly to an eigenstate of the "observable" implemented by the device.

Makes "potentiality" to "actuality".

↑
Hermitian operator.

Postulate 3.4.1 (Measurement Postulate)

A "projective measurement" is described by an observable O which is a Hermitian operator on the state space of the system being observed. Let $O = \sum_{\lambda \in \Lambda} \lambda P_{\lambda}$ be the spectral decomposition of O . The possible outcomes of the measurement are the eigenvalues λ of the observable. When measuring O while the quantum system is in the state $|\psi\rangle$, the probability of getting the result λ is $\text{Pr}_{O,\psi}(\lambda) = \langle \psi | P_{\lambda} | \psi \rangle = \|P_{\lambda} |\psi\rangle\|^2$. If this outcome occurs, the state of the quantum system immediately after the measurement is $\frac{P_{\lambda} |\psi\rangle}{\|P_{\lambda} |\psi\rangle\|}$. $\square$

We have prob space $(\Lambda, \text{Pr}_{O,\psi})$, $\Lambda \subset \mathbb{R}$ since O Hermitian.

Lemma 3.4.2 $E[I_{\Lambda}] = \langle \psi | O | \psi \rangle$.

pf) $E[X] = \sum_{\lambda \in \Lambda} \lambda \text{Pr}(\lambda) = \sum_{\lambda \in \Lambda} \lambda \langle \psi | P_{\lambda} | \psi \rangle = \langle \psi | \sum_{\lambda \in \Lambda} \lambda P_{\lambda} | \psi \rangle = \langle \psi | O | \psi \rangle$.

Def 3.4.3 Let O be an observable of a quantum system with state space H . Suppose that we measure this observable when the system is in the state $|\psi\rangle \in H$. Then the expectation value of this measurement is defined as $\langle \psi | O | \psi \rangle$.

Thm 3.4.4 Assume that we measure an observable O of a quantum system. Let λ be an eigenvalue of O and let $|\psi\rangle$ and $|\psi'\rangle$ be two states of the system that differ only by a global phase factor. Then the probability of measuring λ is the same regardless of whether the system is in the state $|\psi\rangle$ or $|\psi'\rangle$. Also, if the system is in one of the two states and the measurement outcome λ occurs, then the states immediately after the measurement are equal up to the same global phase factor.

pf) $|\psi'\rangle = e^{i\theta} |\psi\rangle \Rightarrow \|P_{\lambda} |\psi'\rangle\| = \|e^{i\theta} P_{\lambda} |\psi\rangle\| = \|P_{\lambda} |\psi\rangle\|$
 $\frac{P_{\lambda} |\psi'\rangle}{\|P_{\lambda} |\psi'\rangle\|} = \frac{e^{i\theta} P_{\lambda} |\psi\rangle}{\|P_{\lambda} |\psi\rangle\|}$

3.4.2

Measuring quantum systems in an orthonormal basis.

$$|\varphi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle, \quad |\alpha_0|^2 + |\alpha_1|^2 = 1.$$

Projections onto $\text{span}\{|0\rangle\}$ or $\text{span}\{|1\rangle\}$ are $P_0 = |0\rangle\langle 0|$, $P_1 = |1\rangle\langle 1|$.

$$O = 0 \cdot P_0 + 1 \cdot P_1 = |1\rangle\langle 1| : \alpha_0 = 0, \alpha_1 = 1.$$

eigenspace $\mathbb{C}|0\rangle$, $\mathbb{C}|1\rangle$.

$$\|P_b|\varphi\rangle\|^2 = |\alpha_b|^2, \quad b \in \{0, 1\}.$$

$$\frac{P_b|\varphi\rangle}{\|P_b|\varphi\rangle\|} = \frac{\alpha_b}{|\alpha_b|} |b\rangle.$$

$$\langle \varphi | O | \varphi \rangle = \langle \alpha_0|0\rangle + \alpha_1|1\rangle | \alpha_1|1\rangle \rangle = |\alpha_1|^2.$$

$$\varphi = |0\rangle \Rightarrow \text{expectation value} = 0$$

$$\varphi = |1\rangle \Rightarrow 1$$

$$\varphi = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \Rightarrow \frac{1}{2}.$$

H : state space, $\beta = (|b_0\rangle, \dots, |b_{k-1}\rangle)$: orthonormal basis.

(ex) $H = \mathbb{C}^n$, $k = 2^n$, $(|0\rangle_n, \dots, |2^n-1\rangle_n)$, projection onto $\mathbb{C}|b_j\rangle : P_j = |b_j\rangle\langle b_j|$

$$O = \sum_{j=0}^{k-1} j P_j = \sum_{j=0}^{k-1} j |b_j\rangle\langle b_j|.$$

matrix repr. $\begin{pmatrix} 0 & & 0 \\ & \ddots & \\ 0 & & k-1 \end{pmatrix}$: Hermitian, k eigenvalues
 $\lambda_j = j$ with eigenspaces $\mathbb{C}|b_j\rangle$, $j \in \mathbb{Z}_k$.

$$|\varphi\rangle := \sum_{j=0}^{k-1} \alpha_j |b_j\rangle, \quad \sum_{j=0}^{k-1} |\alpha_j|^2 = 1.$$

measuring $O \Rightarrow$ eigenvalues $j \in \mathbb{Z}_k$ of O , $\|P_j|\varphi\rangle\|^2 = |\alpha_j|^2$.

$$\frac{P_j|\varphi\rangle}{\|P_j|\varphi\rangle\|} = \frac{\alpha_j}{|\alpha_j|} |b_j\rangle : \text{equiv to } |b_j\rangle.$$

$$\text{Expectation: } \langle \varphi | O | \varphi \rangle = \sum_{j=0}^{k-1} j |\alpha_j|^2.$$

Def 3.4.1 H : state space. $B = (|b_0\rangle, \dots, |b_{k-1}\rangle)$: orthonormal basis.

By measuring the quantum system in the basis B , we mean measuring the observable $O = \sum_{j=0}^{k-1} j |b_j\rangle\langle b_j|$.

(Ex) $|000\rangle \mapsto \frac{|000\rangle + |011\rangle}{\sqrt{2}} = \frac{|0\rangle_3 + |1\rangle_3}{\sqrt{2}}$

Measuring the 3 qubit register in the computational basis means measuring the observable $O = \sum_{j=1}^3 |j\rangle\langle j|$.

$$E = \frac{0}{2} + \frac{3}{2} = \frac{3}{2}$$

3.4.3 Partial Measurements.

"Measure ~~st~~ selectively on specific parts".

(Ex) $H_{AB} = H_A \otimes H_B$, O_A : observable of A , $O_A = \sum_{\lambda \in \Lambda} \lambda P_\lambda$.

$O_{AB} := O_A \otimes I_B$ is an observable of H_{AB} .

$$O_{AB} = \sum_{\lambda \in \Lambda} \lambda P_\lambda \otimes I_B$$

Measuring O_{AB} with state $|\psi\rangle$, outcome is $\lambda \in \Lambda$ with probability $\|P_\lambda \otimes I_B |\psi\rangle\|^2$ and the state after this outcome is $\frac{P_\lambda \otimes I_B |\psi\rangle}{\|P_\lambda \otimes I_B |\psi\rangle\|}$.

(Ex) $H_A = H_B = H$. Goal: measure the first qubit ^{of} Bell state $|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$.

$$O = |1\rangle\langle 1| \otimes I_1$$

$$\text{If } b \in \{0, 1\}, P_b \otimes I_1 |\psi\rangle = \frac{P_b \otimes I_1 |00\rangle + P_b \otimes I_1 |11\rangle}{\sqrt{2}} = \frac{|b\rangle|b\rangle}{\sqrt{2}}$$

Prob that measuring O gives $\frac{1}{\sqrt{2}}$ is $\frac{1}{2}$, state after this measurement outcome is $|b\rangle|b\rangle$.

Alice and Bob.

The situation is easier if the composite system AB is in a separable state $|\psi\rangle = |\varphi\rangle|\xi\rangle$, $|\varphi\rangle, |\xi\rangle \in H$.

For $\lambda \in \Lambda$ we have

$$P_\lambda \otimes I_B |\psi\rangle = P_\lambda \otimes I_B |\varphi\rangle|\xi\rangle = P_\lambda |\varphi\rangle \otimes |\xi\rangle$$

Hence, the probability of measuring λ is

$$\|P_\lambda |\psi\rangle \otimes |\xi\rangle\|^2 = \|P_\lambda |\psi\rangle\|^2 \|\xi\|^2 = \|P_\lambda |\psi\rangle\|^2.$$

This is the probability for obtaining λ when only system A is measured. Also, the state after measuring λ is

$$\frac{P_\lambda |\psi\rangle}{\|P_\lambda |\psi\rangle\|} \otimes |\xi\rangle.$$

This state is the tensor product of the state after measuring system A with the state $|\xi\rangle$ of system B before the measurement.

(Ex) $|\psi\rangle = |x\rangle |x\rangle$.

Measuring only the first qubit in the computational basis of H_1 gives 0 or 1, each with prob $1/2$. If the measurement outcome $b \in \{0, 1\}$ occurs, then the state immediately after the measurement is $|b\rangle |x\rangle$.